# Exploratory Data Analysis — HRI AGRRA Belize

**Coral reef response variables and oceanographic predictors**

HRI Fish Predictor Model

2026-03-15

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```r
library(tidyverse)
library(here)
library(readxl)
library(corrplot)
```

```r
# Output paths
fig_dir   <- here("results", "figures")
table_dir <- here("results", "tables")
dir.create(fig_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(table_dir, recursive = TRUE, showWarnings = FALSE)

# Standardized theme
theme_hri <- function() {
  theme_minimal(base_size = 14) +
    theme(
      plot.title    = element_text(face = "bold", hjust = 0.5, size = 16),
      plot.subtitle = element_text(hjust = 0.5, size = 12, color = "gray50"),
      axis.title.y  = element_text(face = "bold"),
      axis.text.x   = element_text(angle = 45, vjust = 0.5),
      panel.grid.major = element_line(color = "gray85", linetype = "dashed"),
      panel.grid.minor = element_blank(),
      legend.position  = "top"
    )
}
```

## Data Overview

```r
full  <- read.csv(here("results", "tables", "ResponseVariables_full.csv"))
input <- read.csv(here("results", "tables", "ResponseVariables_input.csv"))

full$YEAR <- factor(full$YEAR)
```

The pipeline produces two datasets from AGRRA survey data across three survey years (2018, 2021, 2023) at reef sites along the Belize Barrier Reef:

- **Full dataset**: 448 observations (site x year) with 27 variables covering benthic cover and fish biomass at the family level.
- **Site-averaged dataset**: 294 unique sites with mean response values across years.

### Sample sizes by year

```r
full %>%
  group_by(YEAR) %>%
  summarise(
    n_total    = n(),
    n_benthic  = sum(!is.na(tCORALavg)),
    n_fish     = sum(!is.na(tCommBiomass)),
    .groups = "drop"
```

```r
  ) %>%
  knitr::kable(
    col.names = c("Year", "Total records", "Benthic records", "Fish records")
  )
```

Table 1: Number of survey records per year

| Year | Total records | Benthic records | Fish records |
|---|---|---|---|
| 2018 | 153 | 51 | 104 |
| 2021 | 185 | 94 | 94 |
| 2023 | 110 | 110 | 110 |

### Missingness

Benthic and fish surveys don't always overlap at the same sites, producing NAs when the datasets are merged.

```r
na_pct <- colMeans(is.na(full)) * 100
na_df  <- tibble(variable = names(na_pct), pct_missing = na_pct) %>%
  filter(pct_missing > 0) %>%
  arrange(desc(pct_missing))

ggplot(na_df, aes(x = reorder(variable, pct_missing), y = pct_missing)) +
  geom_col(fill = "#2C3E50") +
  coord_flip() +
  labs(x = NULL, y = "% Missing", title = "Missing Data by Variable") +
  theme_hri()
```

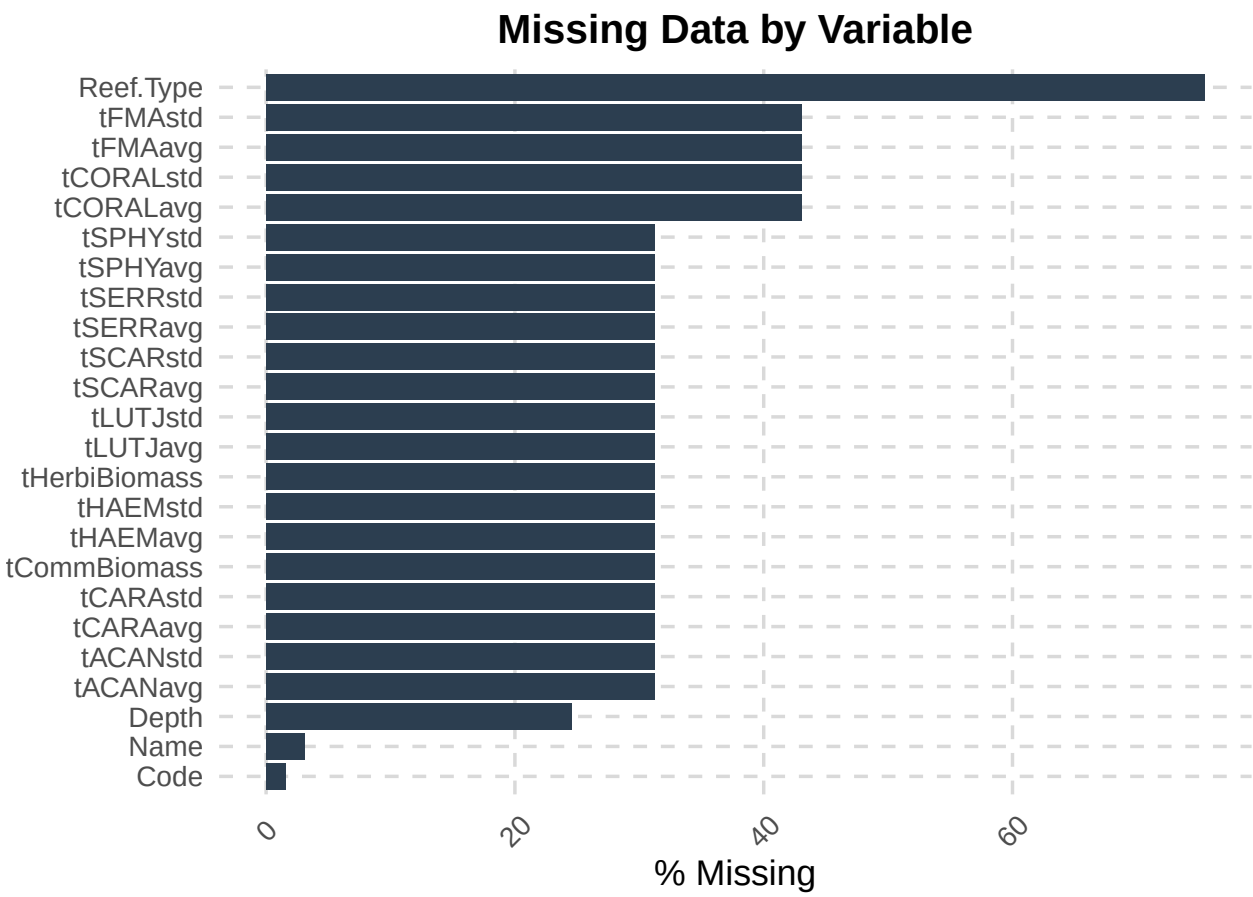



Figure 1: Proportion of missing values per variable

---

## Benthic Cover

### Distributions by year

```
benthic_long <- full %>%
  filter(!is.na(tCORALavg)) %>%
  pivot_longer(
    cols = c(tCORALavg, tFMAavg),
    names_to  = "variable",
    values_to = "value"
  ) %>%
  mutate(variable = recode(variable,
    tCORALavg = "Coral cover (%)",
    tFMAavg   = "Fleshy macroalgae cover (%)"
  ))
```

```r
ggplot(benthic_long, aes(x = YEAR, y = value, fill = variable)) +
  geom_boxplot(alpha = 0.7, outlier.shape = 21,
               outlier.fill = "white", outlier.color = "black") +
  scale_fill_manual(values = c("#1b9e77", "#d95f02")) +
  labs(x = NULL, y = "Cover (%)", fill = NULL,
       title = "Benthic Cover by Year") +
  theme_hri()
```

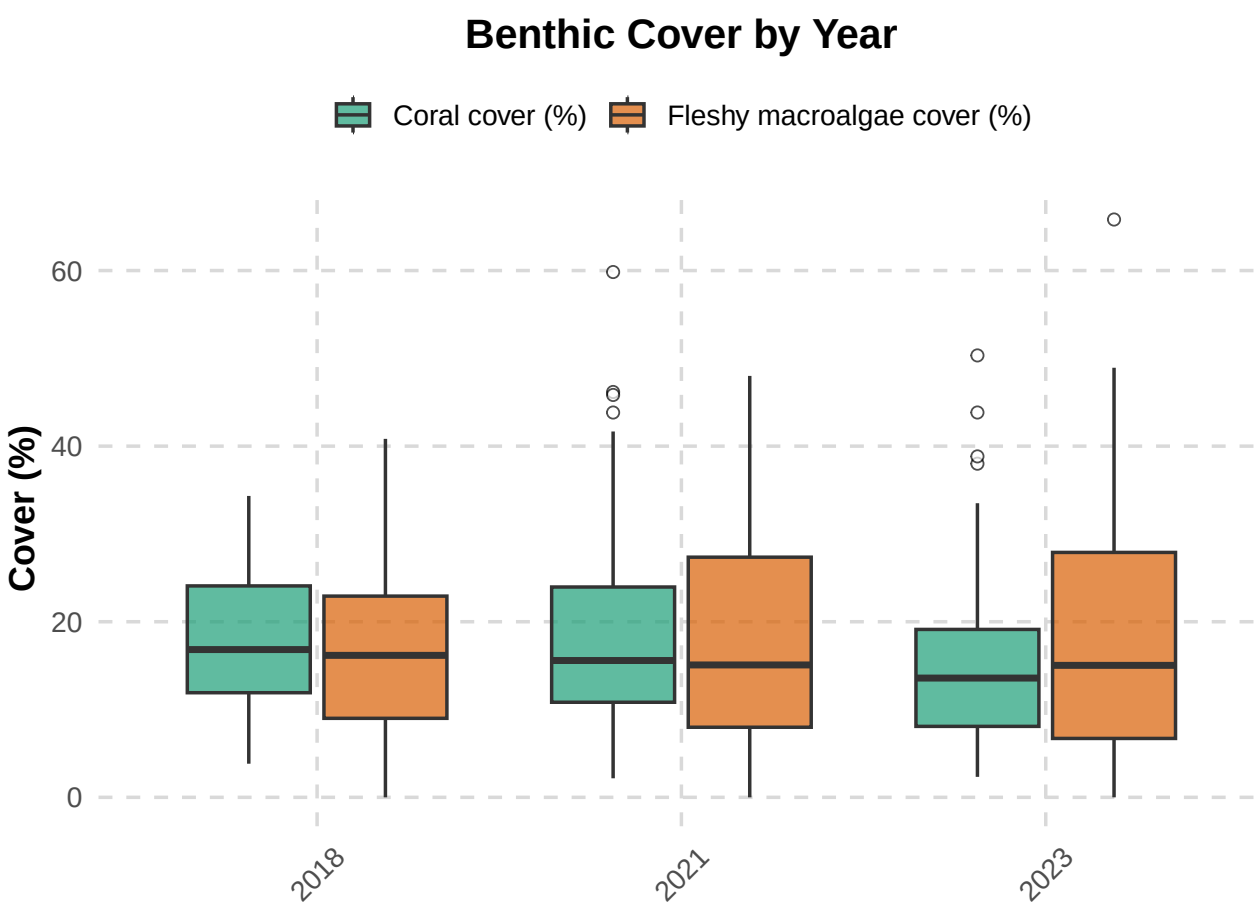



Figure 2: Distribution of coral and macroalgae cover across survey years

### Temporal trends

```r
avg_benthic <- full %>%
  filter(!is.na(tCORALavg)) %>%
  group_by(YEAR) %>%
  summarise(
    coral = mean(tCORALavg),
    algae = mean(tFMAavg),
    n = n(),
```

```
    .groups = "drop"
  )

ggplot(avg_benthic, aes(x = YEAR, group = 1)) +
  geom_line(aes(y = coral, color = "Coral"), linewidth = 1) +
  geom_line(aes(y = algae, color = "Macroalgae"), linewidth = 1) +
  geom_point(aes(y = coral, color = "Coral"), size = 3) +
  geom_point(aes(y = algae, color = "Macroalgae"), size = 3) +
  scale_color_manual(values = c("Coral" = "#1b9e77", "Macroalgae" = "#d95f02")) +
  scale_y_continuous(limits = c(0, 50)) +
  labs(x = "Year", y = "Mean cover (%)", color = NULL,
       title = "Benthic Cover Trends") +
  theme_hri()
```

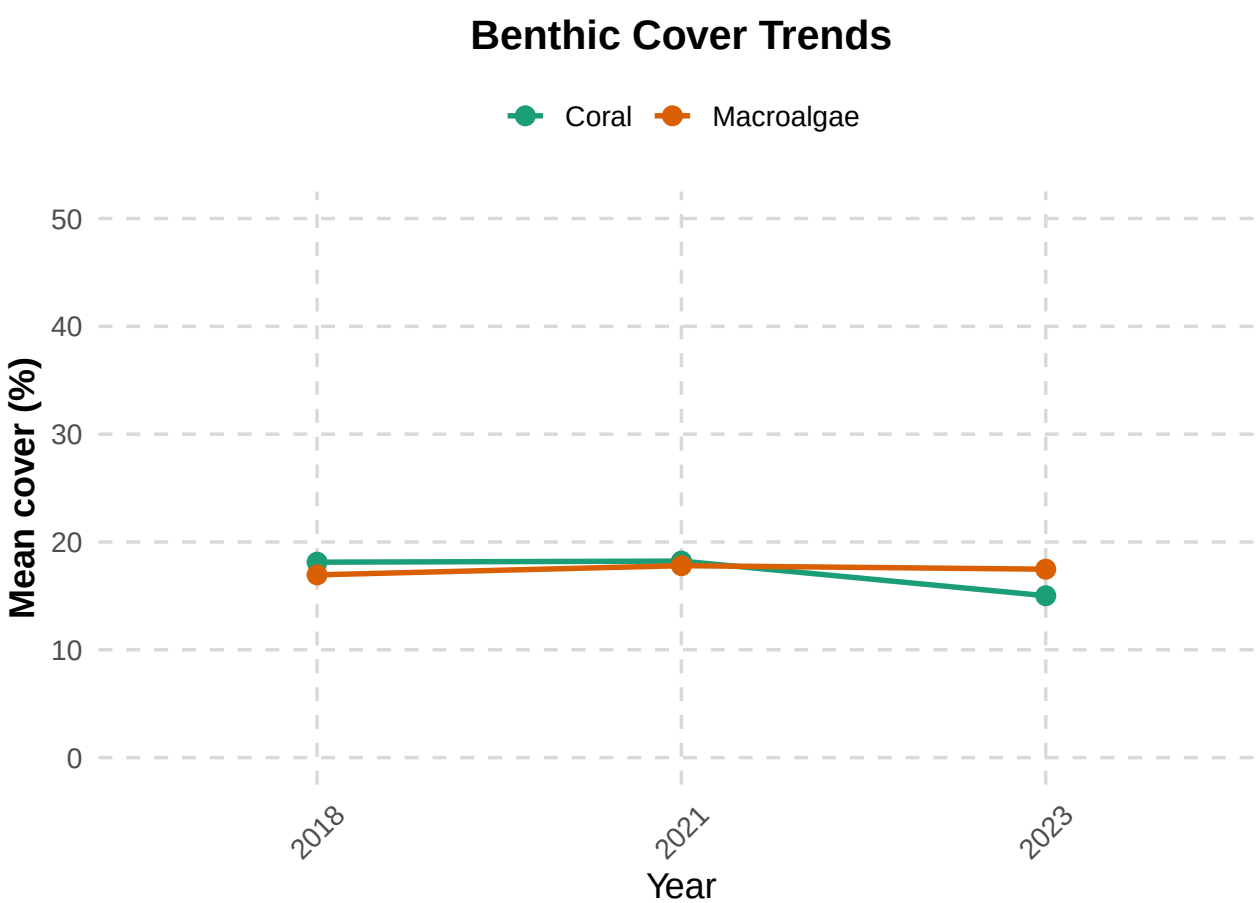



Figure 3: Mean coral and macroalgae cover across survey years

### Coral vs. macroalgae

```
full %>%
  filter(!is.na(tCORALavg)) %>%
```

```
ggplot(aes(x = tFMAavg, y = tCORALavg, color = YEAR)) +
geom_point(alpha = 0.6, size = 2) +
geom_smooth(method = "lm", se = FALSE, linetype = "dashed", linewidth = 0.5) +
scale_color_brewer(palette = "Dark2") +
labs(x = "Fleshy macroalgae cover (%)", y = "Coral cover (%)",
     color = "Year",
     title = "Coral vs. Macroalgae Cover") +
theme_hri()
```

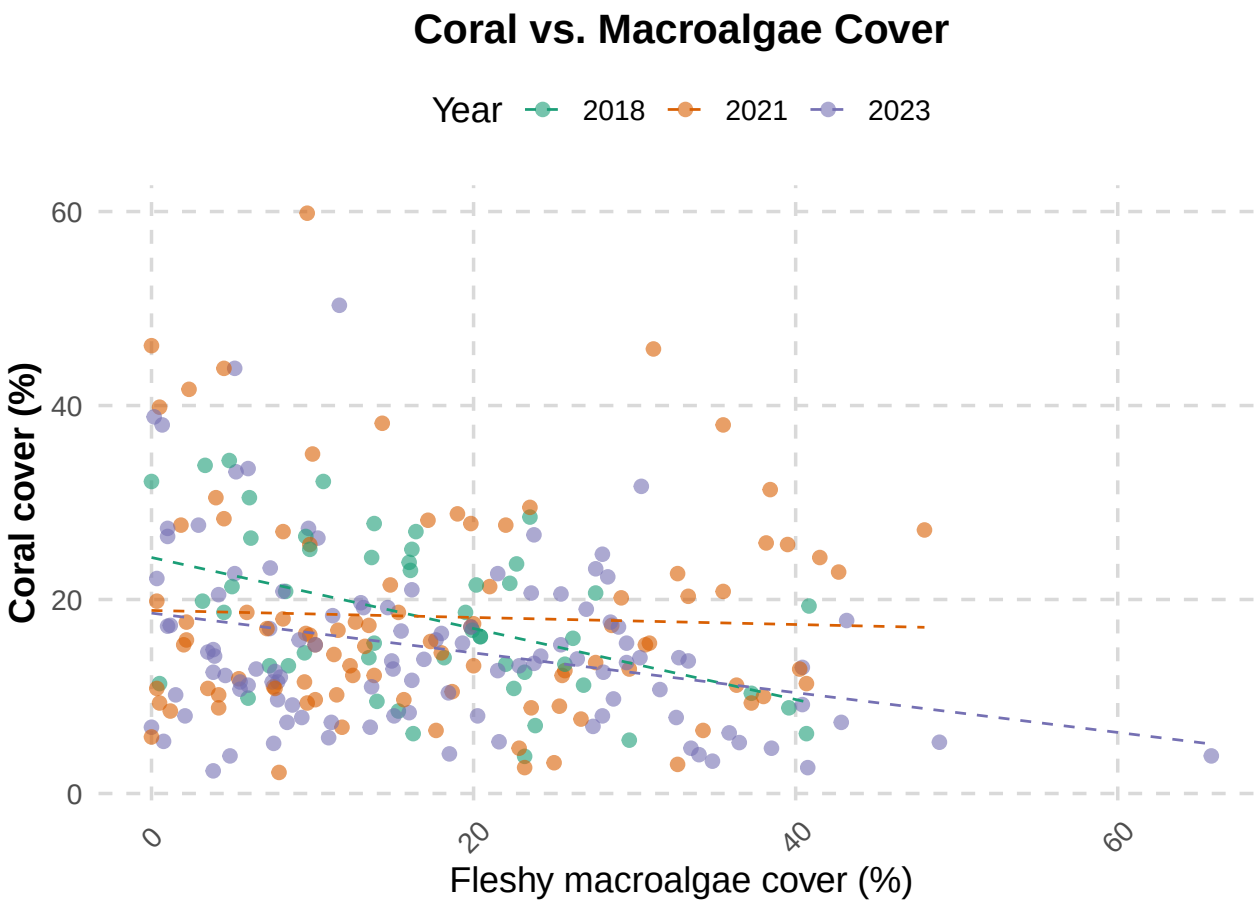



Figure 4: Coral cover vs. macroalgae cover at the site level, colored by year

---

## Fish Biomass

### Commercial fish biomass

Commercial species include snappers (LUTJ), groupers (SERR), jacks (CARA), barracuda (SPHY), and grunts (HAEM). Biomass is in g/100m$^2$.

```r
comm_long <- full %>%
  filter(!is.na(tCommBiomass)) %>%
  select(YEAR, tLUTJavg, tSERRavg, tCARAavg, tSPHYavg, tHAEMavg) %>%
  pivot_longer(-YEAR, names_to = "family", values_to = "biomass") %>%
  mutate(family = recode(family,
    tLUTJavg = "Snappers",
    tSERRavg = "Groupers",
    tCARAavg = "Jacks",
    tSPHYavg = "Barracuda",
    tHAEMavg = "Grunts"
  ))

ggplot(comm_long, aes(x = family, y = biomass, fill = YEAR)) +
  geom_boxplot(alpha = 0.7, outlier.size = 1) +
  scale_y_log10(labels = scales::comma) +
  scale_fill_brewer(palette = "Dark2") +
  labs(x = NULL, y = "Biomass (g/100m², log scale)", fill = "Year",
       title = "Commercial Fish Biomass by Family") +
  theme_hri()
```

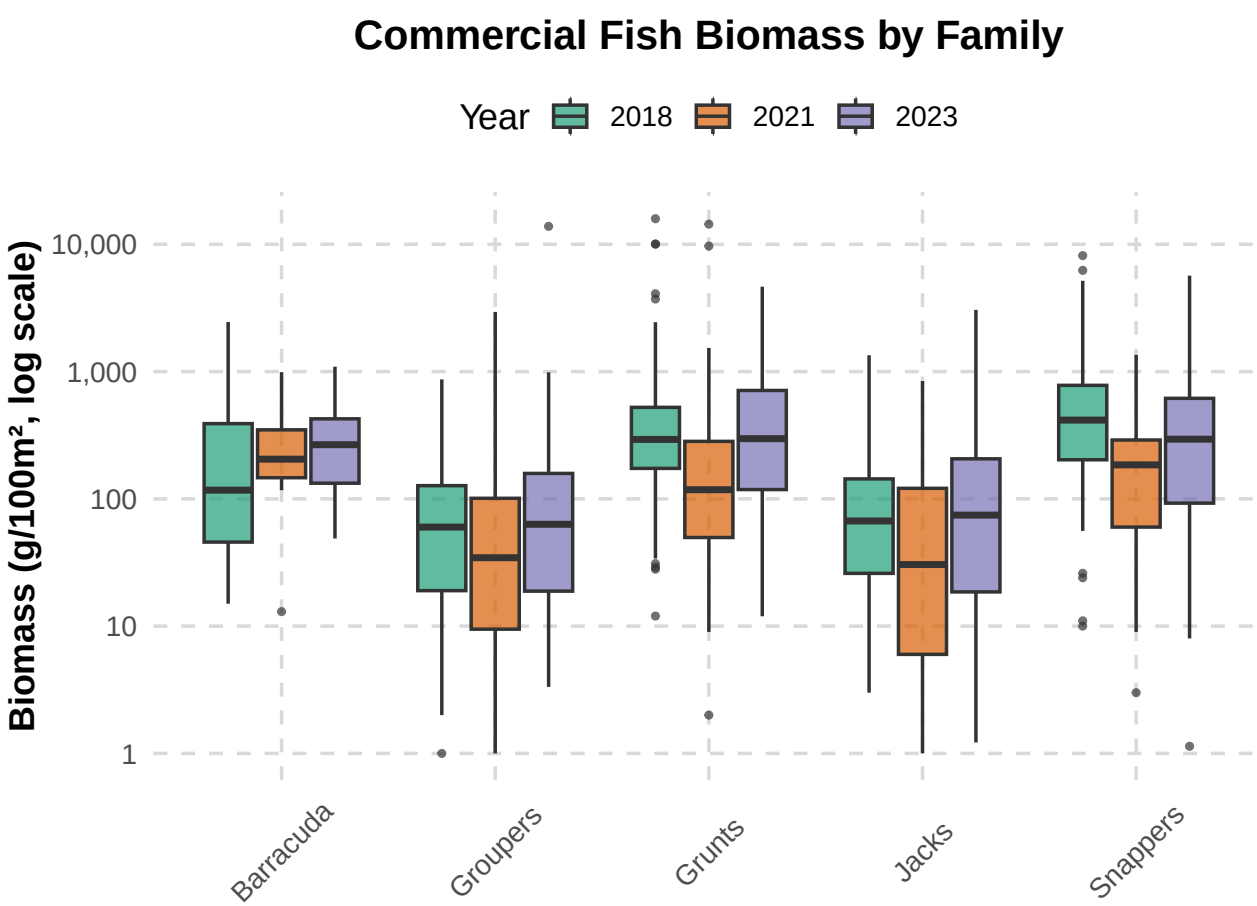



Figure 5: Commercial fish family biomass distributions by year

```r
avg_comm <- full %>%
  filter(!is.na(tCommBiomass)) %>%
  group_by(YEAR) %>%
  summarise(avg = mean(tCommBiomass), n = n(), .groups = "drop")

ggplot(avg_comm, aes(x = YEAR, y = avg, group = 1)) +
  geom_line(color = "#2C3E50", linewidth = 1.2) +
  geom_point(color = "#2C3E50", size = 3, shape = 21,
             fill = "white", stroke = 1.2) +
  labs(x = "Year", y = "Mean biomass (g/100m²)",
       title = "Commercial Fish Biomass Trend") +
  theme_hri()
```

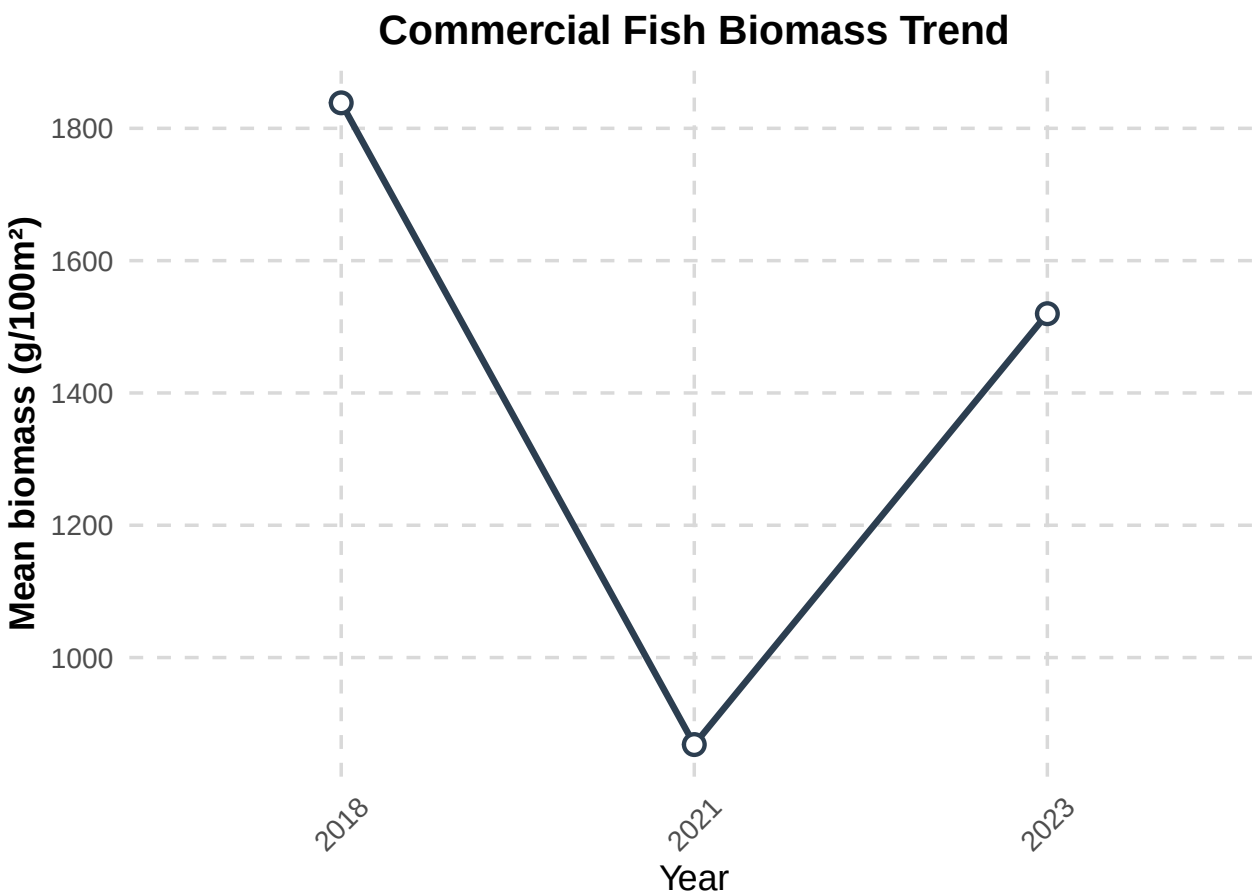



Figure 6: Total commercial fish biomass trend

### Herbivorous fish biomass

Herbivorous species include surgeonfish (ACAN/Acanthuridae) and parrotfish (SCAR/Scaridae).

```r
herb_long <- full %>%
  filter(!is.na(tHerbiBiomass)) %>%
  select(YEAR, tACANavg, tSCARavg) %>%
  pivot_longer(-YEAR, names_to = "family", values_to = "biomass") %>%
  mutate(family = recode(family,
    tACANavg = "Surgeonfish",
    tSCARavg = "Parrotfish"
  ))

ggplot(herb_long, aes(x = family, y = biomass, fill = YEAR)) +
  geom_boxplot(alpha = 0.7, outlier.size = 1) +
  scale_y_log10(labels = scales::comma) +
  scale_fill_brewer(palette = "Dark2") +
  labs(x = NULL, y = "Biomass (g/100m², log scale)", fill = "Year",
       title = "Herbivorous Fish Biomass by Family") +
  theme_hri()
```

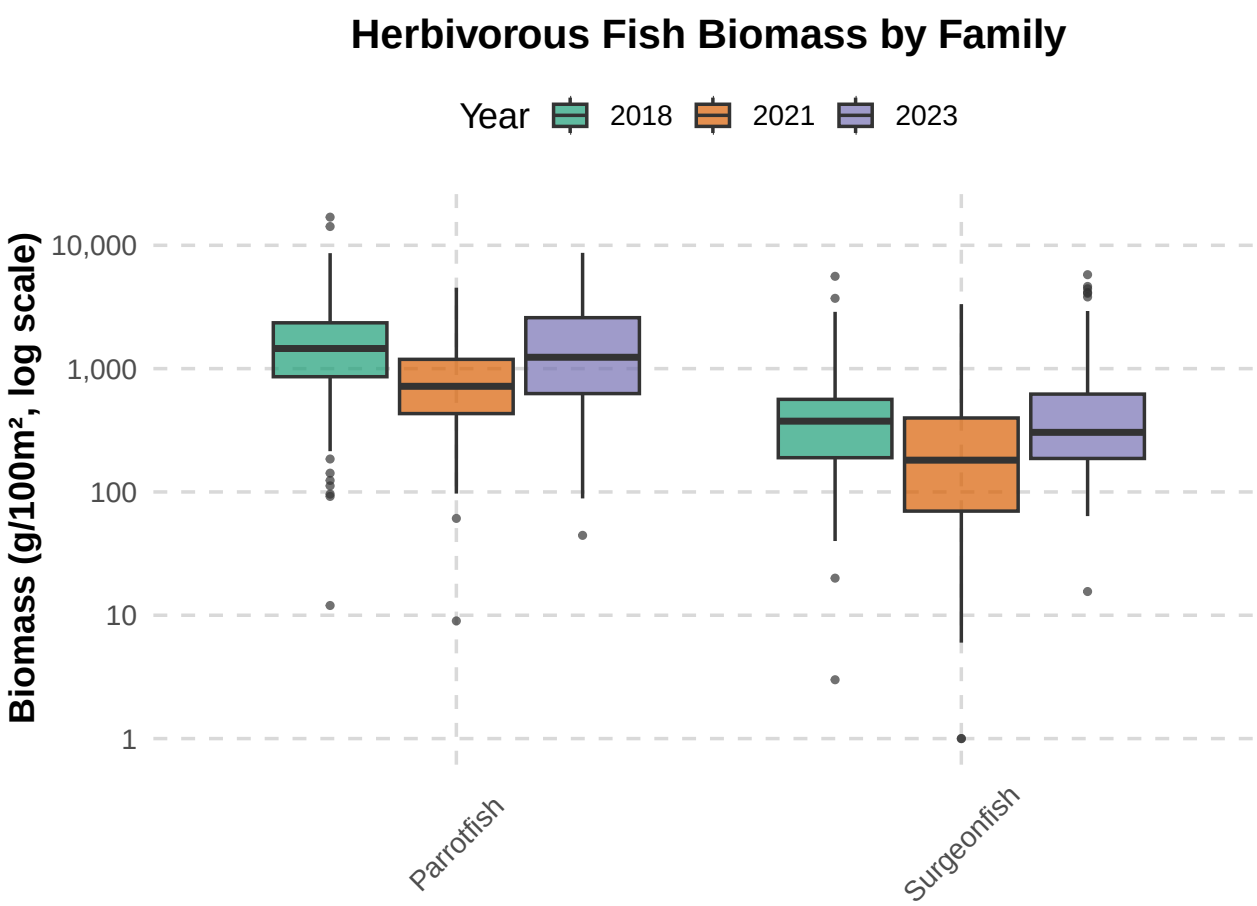



Figure 7: Herbivorous fish family biomass by year

```r
avg_herb <- full %>%
  filter(!is.na(tHerbiBiomass)) %>%
```

```
  group_by(YEAR) %>%
  summarise(avg = mean(tHerbiBiomass), n = n(), .groups = "drop")

ggplot(avg_herb, aes(x = YEAR, y = avg, group = 1)) +
  geom_line(color = "#E74C3C", linewidth = 1.2) +
  geom_point(color = "#E74C3C", size = 3, shape = 21,
             fill = "white", stroke = 1.2) +
  labs(x = "Year", y = "Mean biomass (g/100m²)",
       title = "Herbivorous Fish Biomass Trend") +
  theme_hri()
```

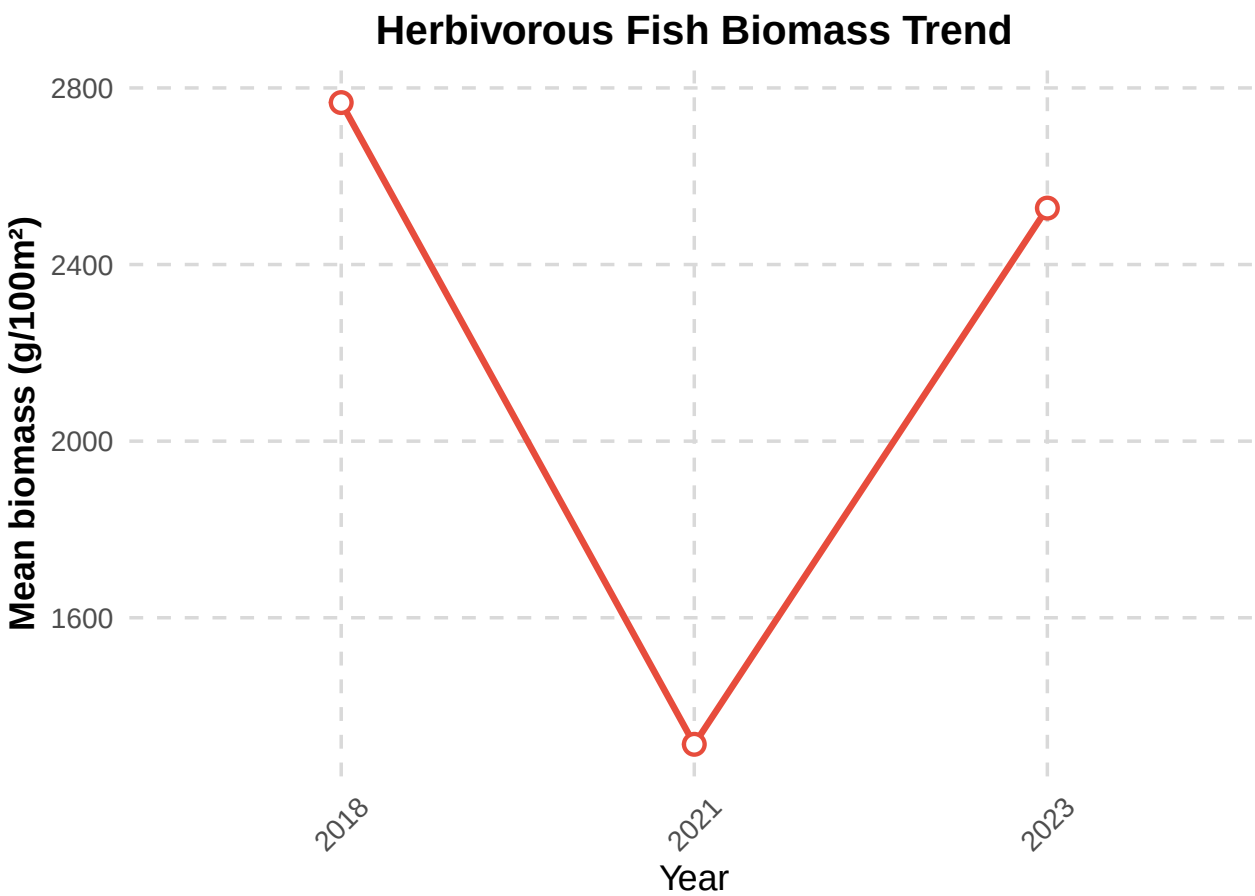



Figure 8: Total herbivorous fish biomass trend

### Combined biomass trends

```
combined <- bind_rows(
  avg_comm %>% mutate(group = "Commercial"),
  avg_herb %>% mutate(group = "Herbivorous")
)
```

```
ggplot(combined, aes(x = YEAR, y = avg, group = group, color = group)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3, shape = 21, fill = "white", stroke = 1.2) +
  scale_color_manual(values = c("Commercial" = "#2C3E50",
                                "Herbivorous" = "#E74C3C")) +
  labs(x = "Year", y = "Mean biomass (g/100m²)", color = NULL,
       title = "Fish Biomass Trends") +
  theme_hri()
```

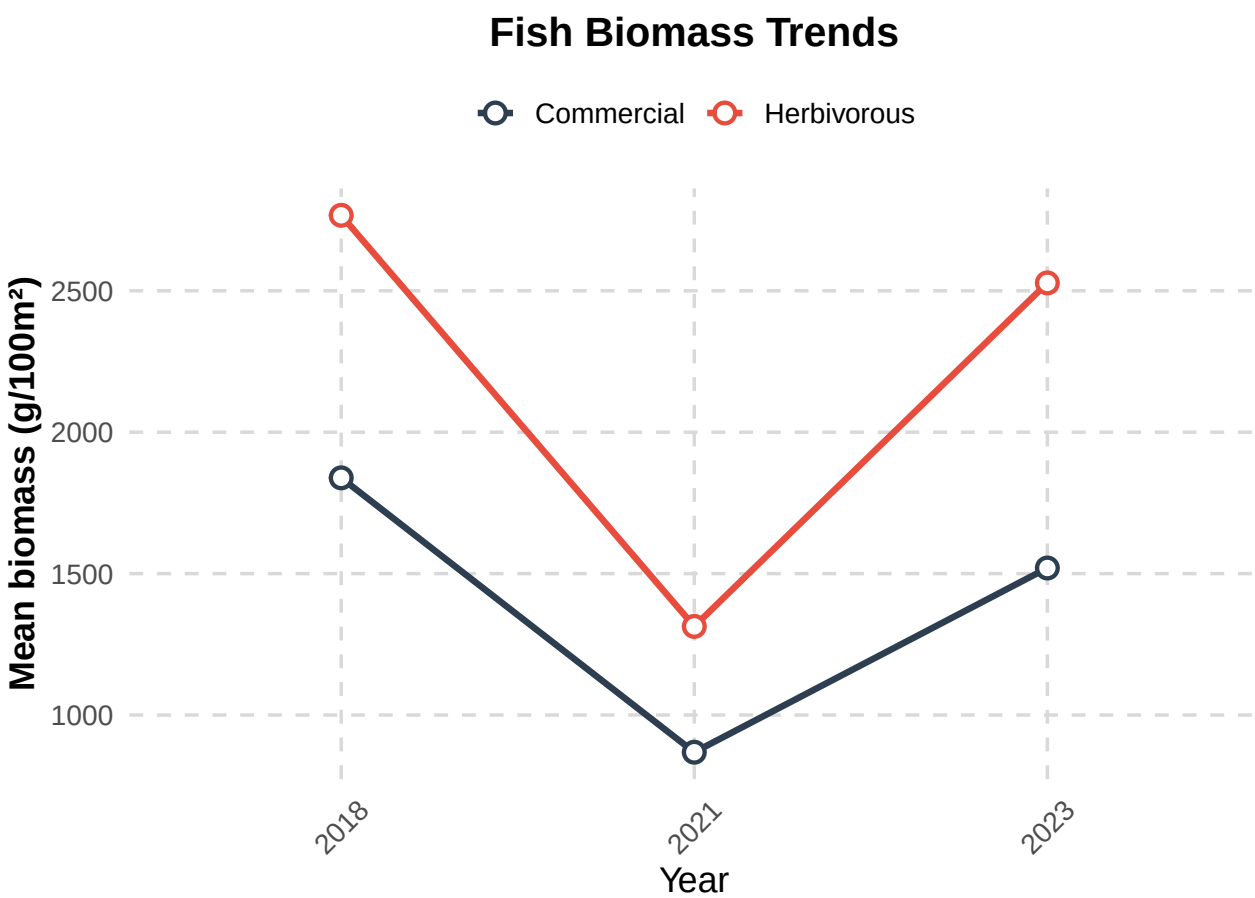



Figure 9: Commercial vs. herbivorous fish biomass trends

---

## Response Variable Relationships

```
resp_vars <- input %>%
  select(tCommBiomass_avg, tHerbiBiomass_avg, tCORALavg_avg, tFMAavg_avg) %>%
  rename(
```

```r
    `Commercial fish` = tCommBiomass_avg,
    `Herbivorous fish` = tHerbiBiomass_avg,
    `Coral cover`  = tCORALavg_avg,
    `Macroalgae`   = tFMAavg_avg
  )

pairs(resp_vars, pch = 16, col = adjustcolor("#2C3E50", alpha.f = 0.5),
      main = "Response Variable Pairs (Site Averages)")
```

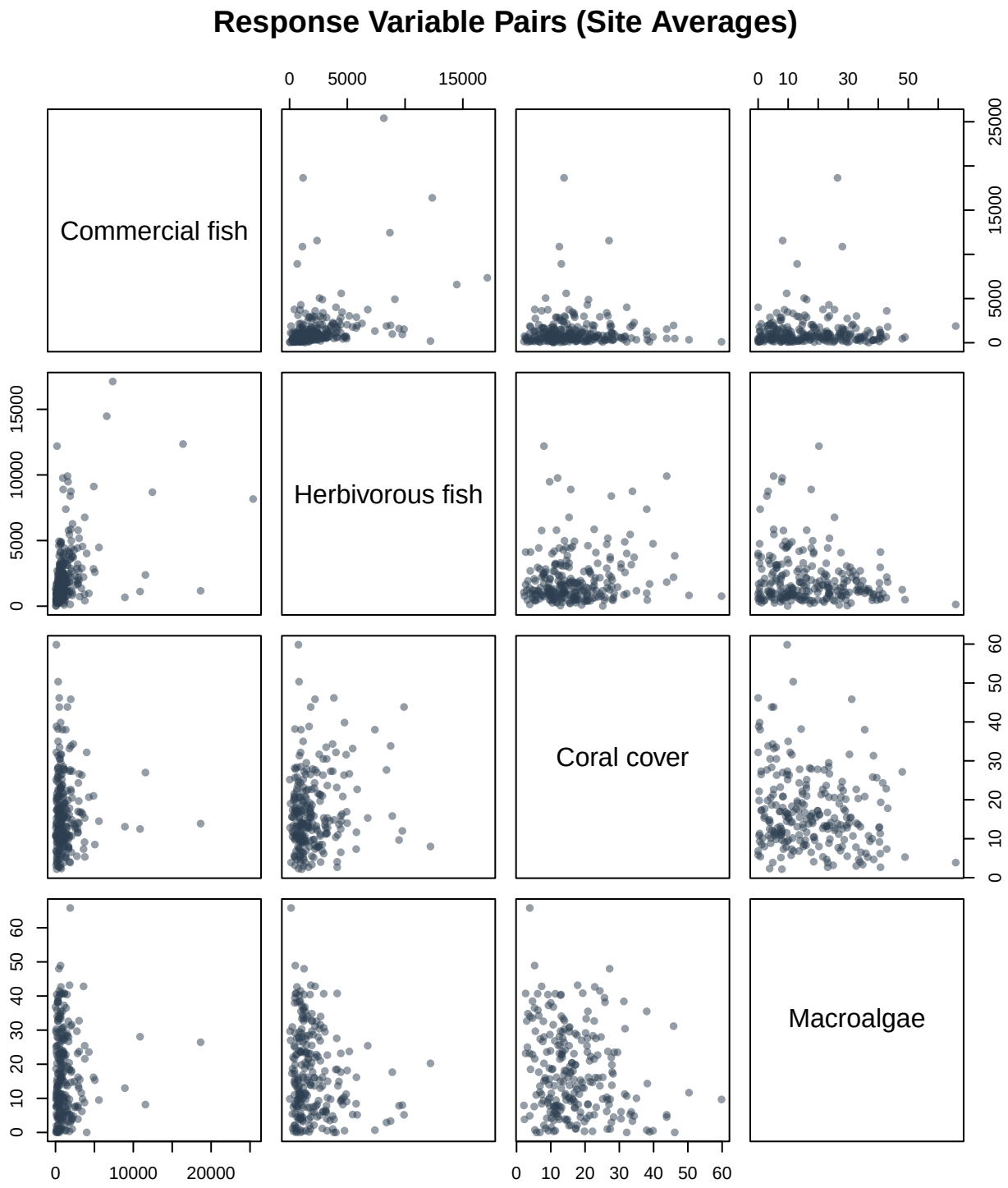



Figure 10: Pairwise relationships among site-averaged response variables

```
resp_complete <- na.omit(resp_vars)
cor_resp <- cor(resp_complete, method = "pearson")

corrplot(cor_resp, method = "number", type = "upper", diag = FALSE,
         number.cex = 1.0, tl.cex = 0.9, tl.col = "black",
```

```
        col = colorRampPalette(c("#E74C3C", "white", "#2C3E50"))(200),
        title = "Response Variable Correlations",
        mar = c(0, 0, 2, 0))
```

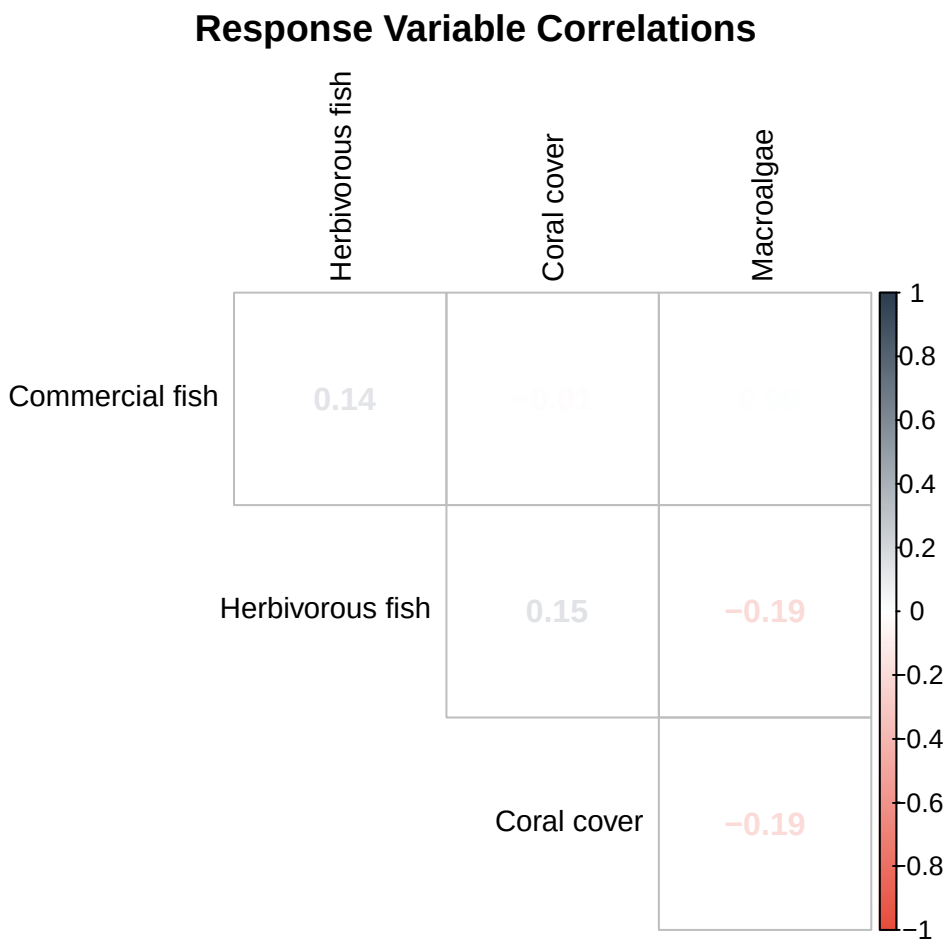
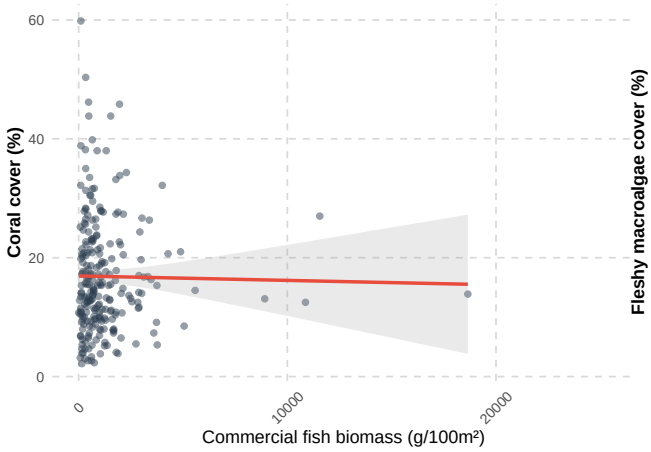



Figure 11: Correlation matrix of response variables

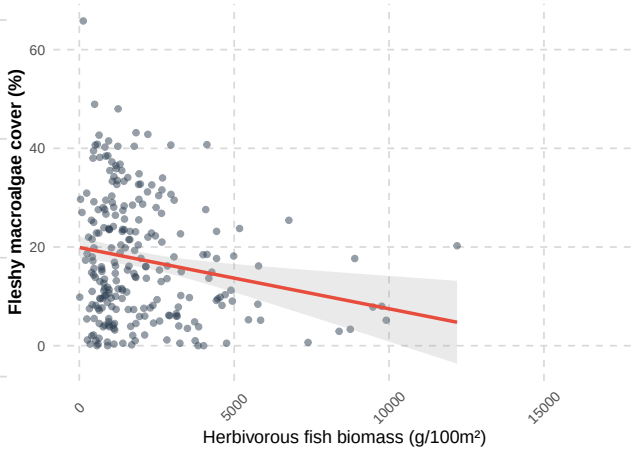
```
p1 <- ggplot(input, aes(x = tCommBiomass_avg, y = tCORALavg_avg)) +
  geom_point(alpha = 0.5, color = "#2C3E50") +
  geom_smooth(method = "lm", color = "#E74C3C", se = TRUE, alpha = 0.2) +
  labs(x = "Commercial fish biomass (g/100m²)",
       y = "Coral cover (%)") +
  theme_hri()

p2 <- ggplot(input, aes(x = tHerbiBiomass_avg, y = tFMAavg_avg)) +
  geom_point(alpha = 0.5, color = "#2C3E50") +
  geom_smooth(method = "lm", color = "#E74C3C", se = TRUE, alpha = 0.2) +
  labs(x = "Herbivorous fish biomass (g/100m²)",
       y = "Fleshy macroalgae cover (%)") +
  theme_hri()
```

```
p1
p2
```




(a) Key ecological relationships




(a) Key ecological relationships

---

## Spatial Distribution

```
input_map <- input %>% filter(!is.na(tCORALavg_avg))

ggplot(input_map, aes(x = Longitude, y = Latitude, color = tCORALavg_avg)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis_c(option = "C", name = "Coral\ncover (%)") +
  labs(title = "Survey Sites - Coral Cover",
       x = "Longitude", y = "Latitude") +
  coord_quickmap() +
  theme_hri()
```

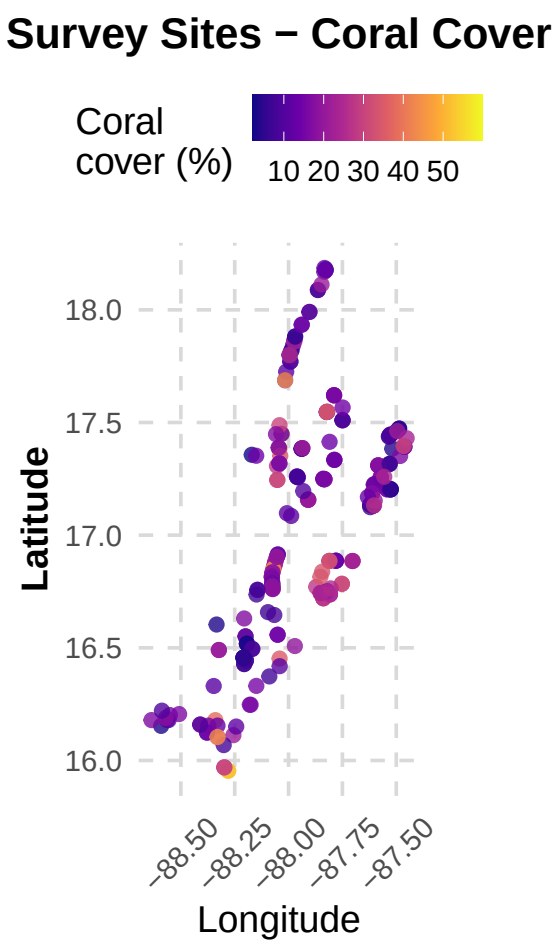



Figure 14: Survey site locations along the Belize Barrier Reef, colored by mean coral cover

```
input_fish <- input %>% filter(!is.na(tHerbiBiomass_avg))

ggplot(input_fish, aes(x = Longitude, y = Latitude, color = tHerbiBiomass_avg)) +
  geom_point(size = 2, alpha = 0.8) +
  scale_color_viridis_c(option = "D", name = "Herb. fish\n(g/100m²)") +
  labs(title = "Survey Sites - Herbivorous Fish Biomass",
       x = "Longitude", y = "Latitude") +
  coord_quickmap() +
  theme_hri()
```

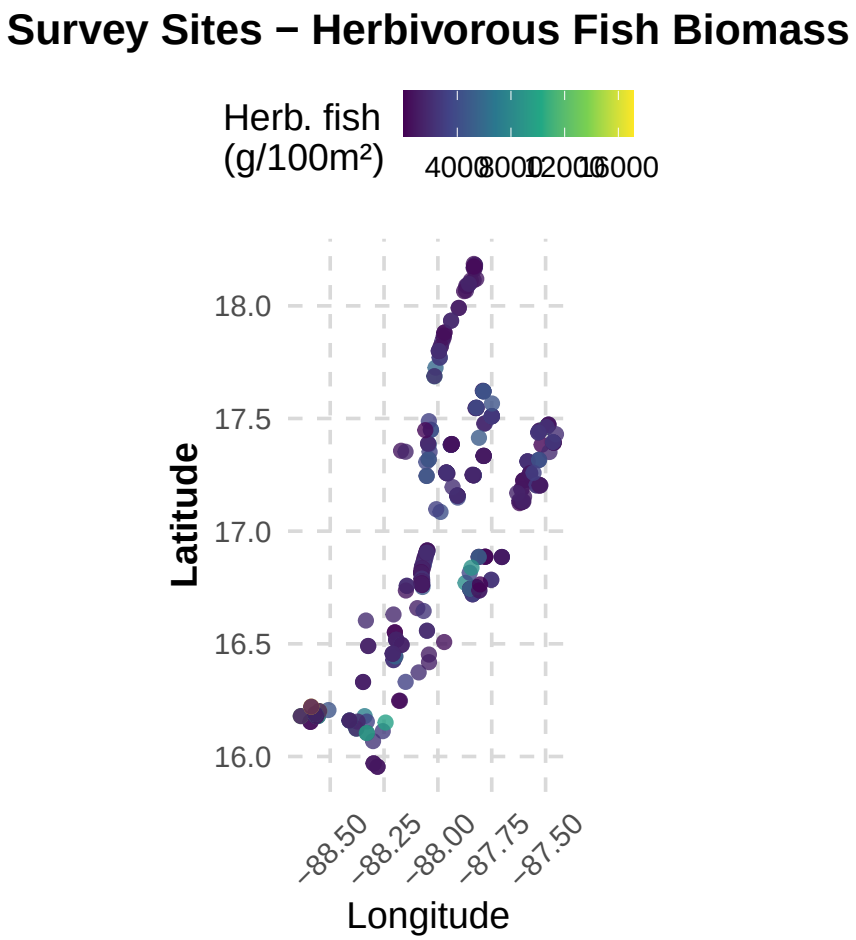



Figure 15: Survey sites colored by mean herbivorous fish biomass

---

## Predictor Variable Correlation

The oceanographic/bathymetric predictor variables were extracted from ArcGIS rasters at 10m resolution (TNC Bathymetry) and joined to survey sites.

```
ext_path <- here("data", "external", "HRI_all_variables.csv")
stopifnot(
  "External data file missing - see data/external/README.md" =
    file.exists(ext_path)
)

hri <- read.csv(ext_path)

# Rename to descriptive labels
hri <- hri %>%
```

```r
  rename(
    lon           = Longitude_,
    lat           = Latitude_y,
    coral         = Coral_cove,
    algae         = Algae_cove,
    fish_comm     = Commercial,
    fish_herb     = Herbivorou,
    aspect        = aspect_10m,
    aspect_1      = aspect_1_1,
    aspect_2      = aspect_1_2,
    aspect_s_curv = aspect_s_1,
    bathy         = depth
  )
```

```r
# Keep only oceanographic/terrain predictors (drop response + coordinate cols)
response_cols <- c("FID", "lon", "lat", "coral", "algae", "fish_comm", "fish_herb")

predictors <- hri %>%
  select(-all_of(response_cols)) %>%
  mutate(across(where(is.numeric), ~ifelse(. == -9999, NA, .)))

pred_complete <- na.omit(predictors)
cat("Predictor variables:", ncol(pred_complete), "\n")
```

```
Predictor variables: 13
```

```r
cat("Complete cases:", nrow(pred_complete), "of", nrow(predictors), "\n")
```

```
Complete cases: 267 of 267
```

### Full correlation matrix

```r
cor_all <- cor(pred_complete, use = "complete.obs", method = "pearson")

corrplot(cor_all, method = "number", type = "upper", diag = FALSE,
         number.cex = 0.6, tl.cex = 0.7, tl.col = "black",
         title = "All Predictor Correlations",
         mar = c(0, 0, 2, 0))
```

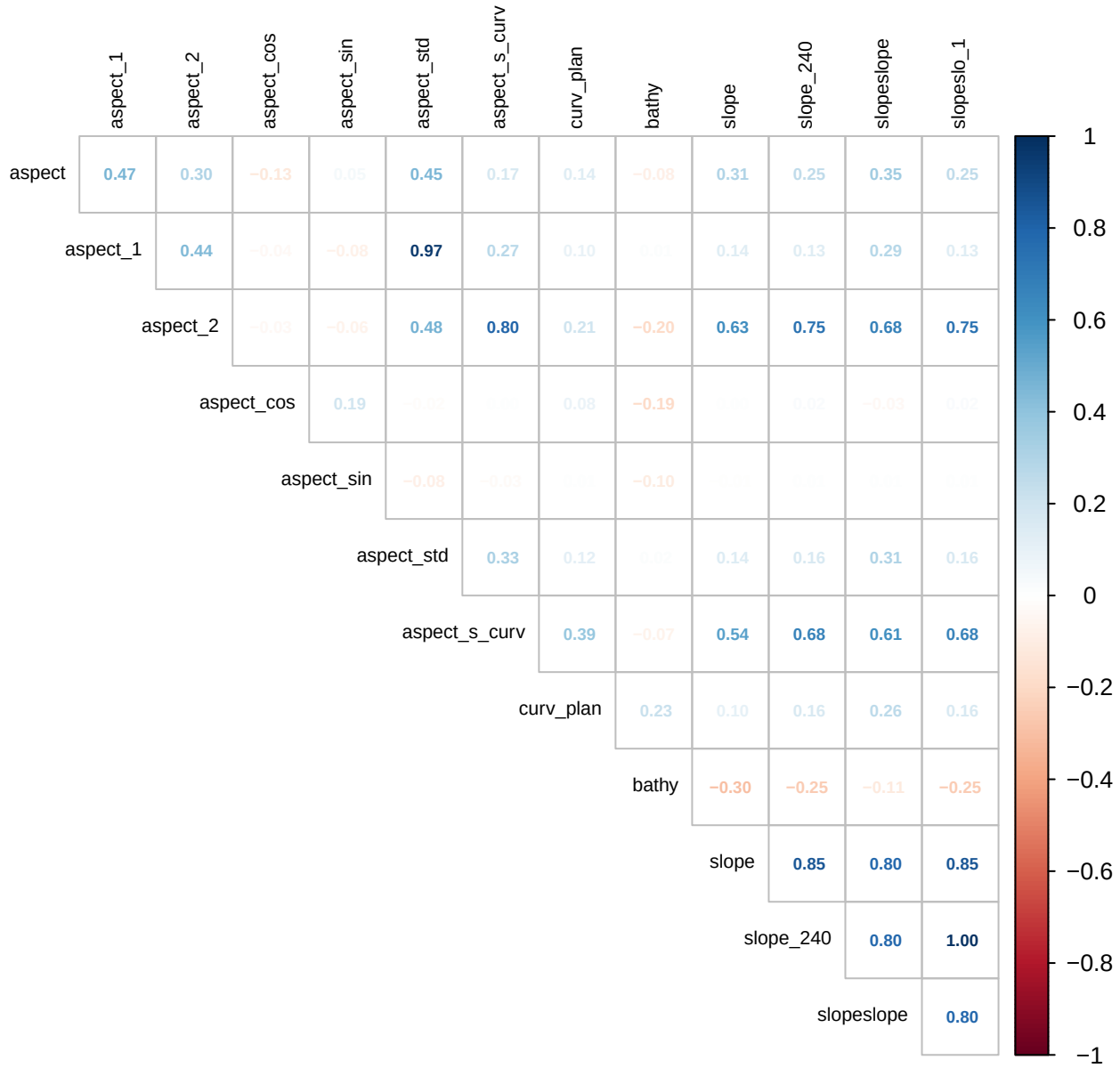



Figure 16: Pearson correlation among all predictor variables

```
# Save to file
png(file.path(fig_dir, "correlation_all_vbles.png"),
    width = 8, height = 8, units = "in", res = 300)
corrplot(cor_all, method = "number", type = "upper", diag = FALSE,
         number.cex = 0.6, tl.cex = 0.7, tl.col = "black")
dev.off()
```

```
pdf
  2
```

### Identify collinear variables

Variables with |r| > 0.7 should be removed before modeling to reduce multicollinearity.

```r
high_cor <- which(abs(cor_all) > 0.7 & upper.tri(cor_all), arr.ind = TRUE)

if (nrow(high_cor) > 0) {
  cor_pairs <- tibble(
    var1 = rownames(cor_all)[high_cor[, 1]],
    var2 = colnames(cor_all)[high_cor[, 2]],
    r    = cor_all[high_cor]
  ) %>%
    arrange(desc(abs(r)))

  knitr::kable(cor_pairs, digits = 3)
} else {
  cat("No variable pairs exceed |r| > 0.7\n")
}
```

Table 2: Variable pairs with |r| > 0.7

| var1 | var2 | r |
|---|---|---|
| slope_240 | slopeslo_1 | 1.000 |
| aspect_1 | aspect_std | 0.970 |
| slope | slope_240 | 0.851 |
| slope | slopeslo_1 | 0.851 |
| aspect_2 | aspect_s_curv | 0.802 |
| slope_240 | slopeslope | 0.799 |
| slopeslope | slopeslo_1 | 0.799 |
| slope | slopeslope | 0.797 |
| aspect_2 | slope_240 | 0.746 |
| aspect_2 | slopeslo_1 | 0.746 |

```r
# Remove variables identified as redundant
exclude_vars <- c("curv_plan", "slopeslo_1", "slope_240")
pred_reduced <- pred_complete %>%
  select(-any_of(exclude_vars))

cat("Retained predictors:", ncol(pred_reduced), "\n")
```

```
Retained predictors: 10
```

```r
cat("Removed:", paste(intersect(exclude_vars, names(pred_complete)), collapse = ", "), "\n")
```

```
Removed: curv_plan, slopeslo_1, slope_240
```

### Reduced correlation matrix

```r
cor_reduced <- cor(pred_reduced, use = "complete.obs", method = "pearson")

corrplot(cor_reduced, method = "number", type = "upper", diag = FALSE,
         number.cex = 0.7, tl.cex = 0.8, tl.col = "black",
         title = "Retained Predictor Correlations",
         mar = c(0, 0, 2, 0))
```

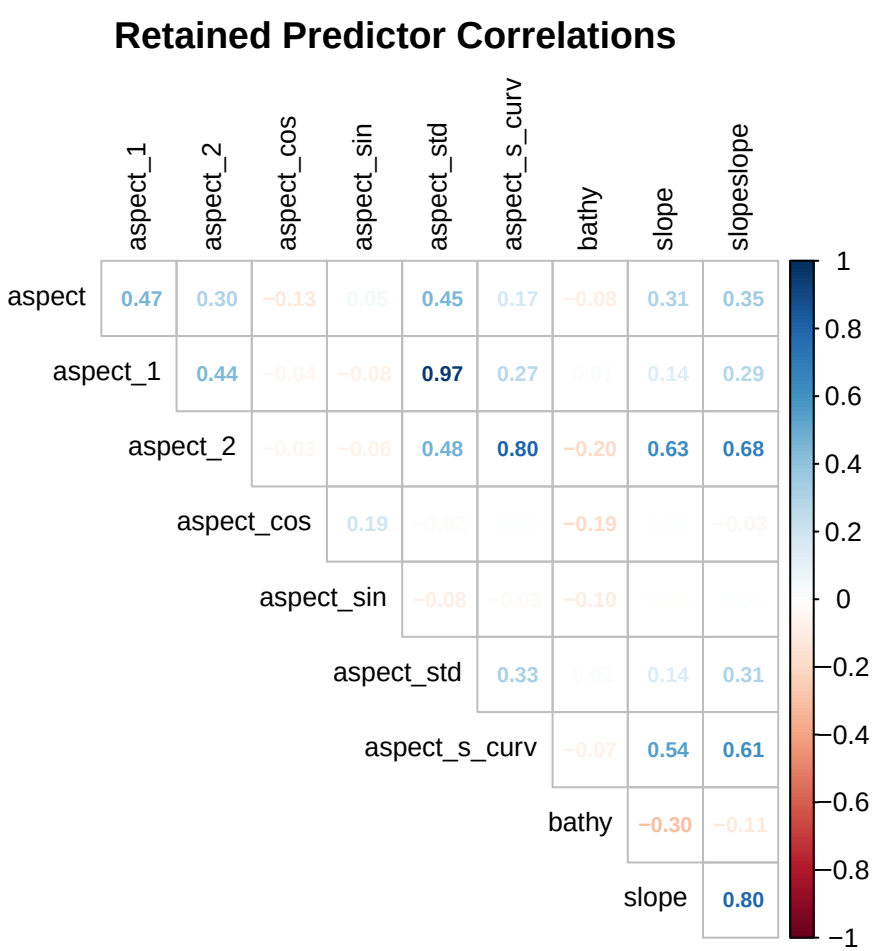



Figure 17: Correlation matrix after removing collinear variables

```r
png(file.path(fig_dir, "correlation_final.png"),
    width = 7, height = 7, units = "in", res = 300)
```

```r
corrplot(cor_reduced, method = "number", type = "upper", diag = FALSE,
         number.cex = 0.7, tl.cex = 0.8, tl.col = "black")
dev.off()
```

```
pdf
  2
```

### Predictor distributions

```r
pred_long <- pred_reduced %>%
  pivot_longer(everything(), names_to = "variable", values_to = "value")

ggplot(pred_long, aes(x = value)) +
  geom_histogram(bins = 30, fill = "#2C3E50", alpha = 0.7, color = "white") +
  facet_wrap(~variable, scales = "free", ncol = 3) +
  labs(x = NULL, y = "Count",
       title = "Predictor Variable Distributions") +
  theme_hri() +
  theme(axis.text.x = element_text(angle = 0, size = 8))
```

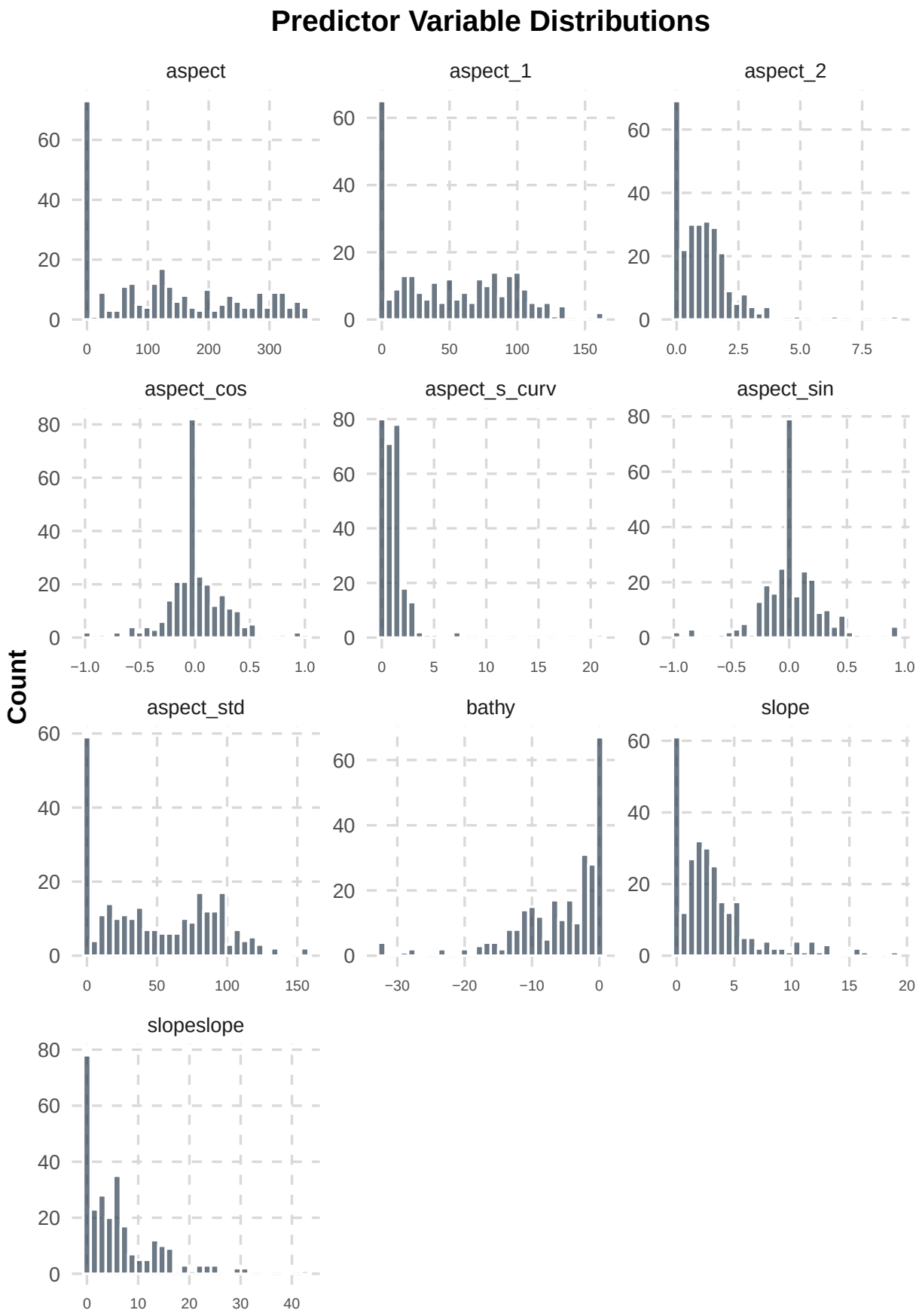



Figure 18: Distributions of retained predictor variables

---

## Export model-ready data

```
data_model <- hri %>%
  select(-any_of(exclude_vars))

write.csv(data_model,
          file = file.path(table_dir, "bz_data_model.csv"),
          row.names = FALSE)
cat("Exported:", nrow(data_model), "rows x", ncol(data_model), "cols\n")
```

```
Exported: 267 rows x 17 cols
```

---

## Summary

| Metric | Value |
|---|---|
| Survey years | 2018, 2021, 2023 |
| Total observations | 448 |
| Unique sites | 294 |
| Mean coral cover | 16.8% |
| Mean macroalgae cover | 17.5% |
| Mean commercial fish biomass | 1429 g/100m$^2$ |
| Mean herbivorous fish biomass | 2238 g/100m$^2$ |
| Predictor variables (retained) | 10 |